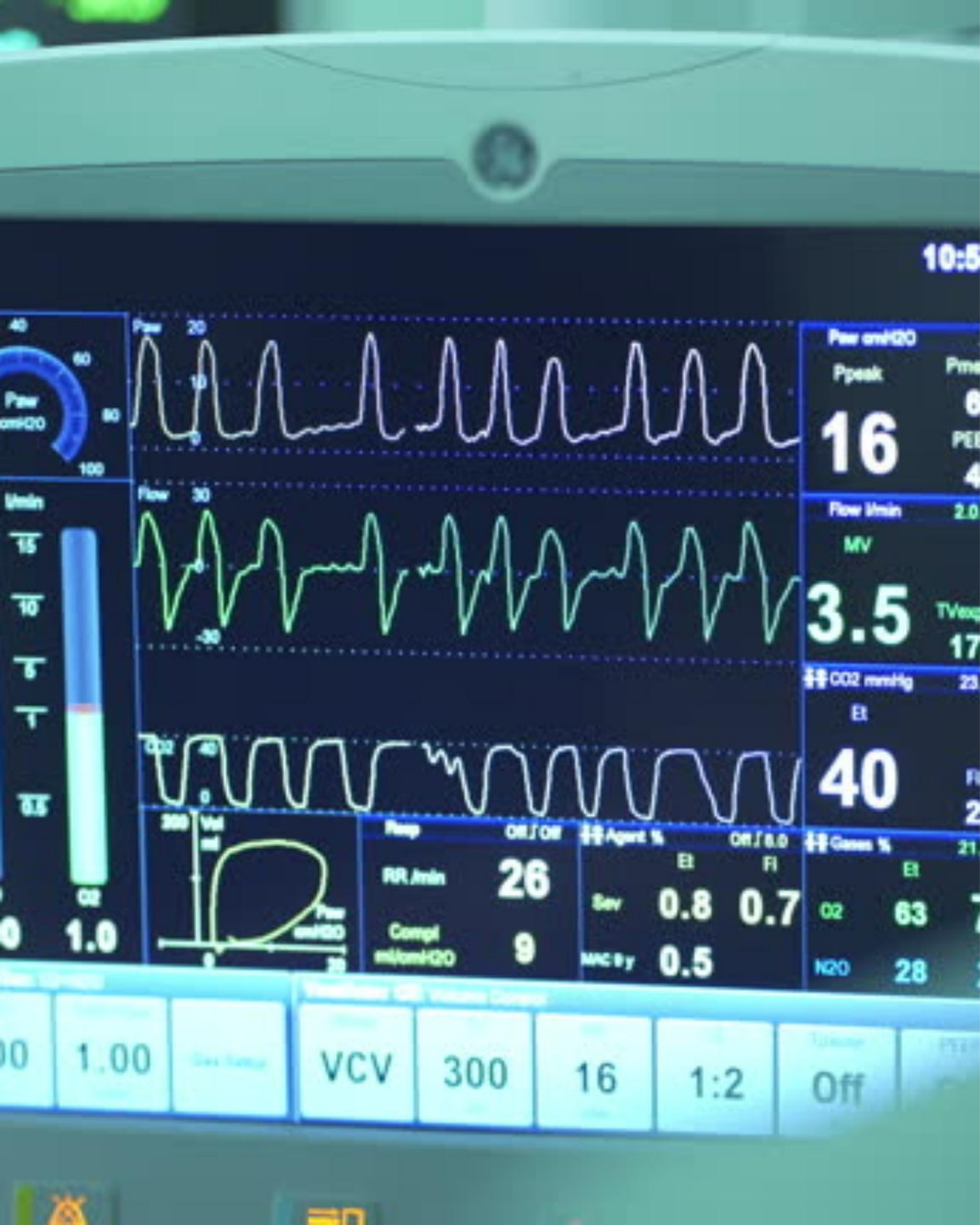


HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

Sepsis & Organ Stress Digital Twin

Nikesh Kumar

Nagaland University



The ICU Challenge

- ⚠ **Rapid Onset:** High risk of abrupt multi-organ failure.
- 🕒 **Late Alarms:** Routine monitors trigger after clinical onset.
- ⌚ **Narrow Window:** Minimal time left for life-saving interventions.

AI-Powered Solution



Data Fusion

Combines wearable
vitals with EHR lab
records.



ML Engine

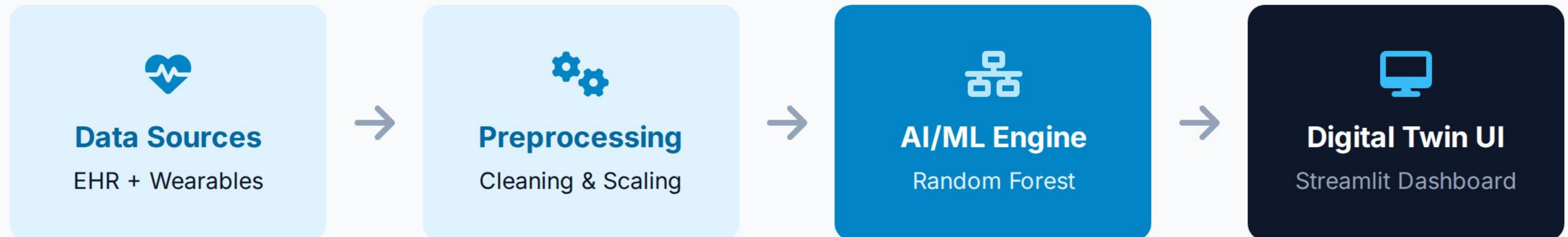
Spots subtle stress
patterns autonomously.



Early Warning

Predicts sepsis onset 6-
8 hours in advance.

System Architecture



Technical Stack



Python

Core development language.



Scikit-Learn

Random Forest classifier.



Pandas & NumPy

Data manipulation & processing.



Streamlit

Real-time interactive dashboard.

Clinical Tracking Metrics

92%

Sepsis Risk Score

Live probability of onset.

Kidney

Stress Index

Monitors Creatinine levels.

Lungs

Respiratory Index

Tracks SpO2 & strain.

Thank You

Nikesh Kumar | Nagaland University

github.com/nikeshyadavp7488-ctrl/sepsis-digital-twin-challenge

Image Sources



https://media.istockphoto.com/id/2229344492/video/4k-medical-data-monitor-in-operation-room.jpg?s=640x640&k=20&c=7ol__LoMMS_JLqeedVv8nZ5LBklbLutXpqsYlsZsi0=

Source: www.istockphoto.com